

VORLAN NASYWAN FATAHILLAH

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Bandung, Indonesia

Fresh graduate in Informatics from Institut Teknologi Nasional with a GPA of 3.68/4.00, with a strong foundation in System Design, Database Architecture, and Software Development Life Cycle (SDLC). Hands-on experience in web development, backend development, machine learning, cloud computing, computer networking, and Internet of Things. Experienced in developing web-based and cloud-based applications using Laravel, Flask, Node.js, Spring Boot, Go, and AWS, as well as implementing machine learning solutions using Python. Experienced in technical assistance, system troubleshooting, software testing, and project-based development. Strong problem-solving, teamwork, communication, mentoring, and leadership abilities.

Education

Institut Teknologi Nasional - Bandung, Indonesia 2022 - 2026

Bachelor Degree in Informatics, GPA: 3.68/4.00

- Bachelor's Thesis: Designed and developed a web load distribution system using an adaptive load balancer with an ARIMA prediction model and Adaptive Weight Adjustment Model, incorporating actual and predicted server CPU utilization for dynamic traffic distribution.
- Informatics Projects: Developed and implemented various projects in web development, machine learning, cloud computing, computer networking, and Internet of Things (IoT), including applications using Laravel, Flask, Node.js, Spring Boot, Python, and AWS.

Work Experiences

Laboratory of Informatics Institut Teknologi Nasional - Bandung, Indonesia 2025 - 2026

Informatics Laboratory Assistant 2025 – 2026

- Assisted practical sessions and provided technical guidance for Computer Networks, Object-Oriented Programming, Digital Image Processing, Internet of Things, Robotics and Multimedia courses.
- Assisted students in understanding technical concepts and solving programming and technical problems during practical sessions.
- Performed software and network troubleshooting to support smooth laboratory activities.
- Prepared, configured, and tested software and network infrastructure required for laboratory sessions.

Project Experiences

Institut Teknologi Nasional – Bandung, Indonesia 2025

Cloud-Based Health Website (AWS)

- Architected and developed a cloud-based healthcare website using AWS services, including EC2, S3, RDS, SNS, API Gateway, DynamoDB, Lambda, VPC, and IAM.
- Implemented automatic product image upload to Amazon S3 through a Go-based backend and Express + EJS frontend.
- Utilized MySQL RDS for product and transaction data management.
- Implemented JWT Authentication to secure user access.
- Developed a basic cloud deployment pipeline to support application reliability and scalability.
- Designed the relational database schema (MySQL RDS) and system architecture flow for end-to-end user transactions

Institut Teknologi Nasional – Bandung, Indonesia 2024

Computer Networking Project

- Designed and implemented various network topologies based on a hotel floor plan.
- Performed IP address allocation based on network requirements.
- Configured routers, switches, and access points to support stable network communication.
- Conducted end-to-end network testing to verify connectivity and system performance.

Institut Teknologi Nasional – Bandung, Indonesia 2024-2025

Machine Learning and Deep Learning Projects

- Collected and prepared datasets from Kaggle for machine learning applications.
- Developed a machine learning model for flood prediction in Central Java.
- Developed machine learning and deep learning models for rice leaf disease classification.
- Developed a vehicle speed detection system using YOLO and Python.

Institut Teknologi Nasional – Bandung, Indonesia 2023-2026
Web Development Projects: Laravel, Node.js, Flask, and Spring Boot

- Developed various web applications using Laravel, Node.js, Flask, and Spring Boot.
- Designed application concepts, features, and workflows based on system requirements.
- Developed a Game Store website and an e-commerce website for public use.
- Developed a Flask-based document search application for rice disease documents using VSM, TF-IDF, and NLP.
- Developed an OSIS management system for SMAN 10 Bandung as part of a community service project.

Institut Teknologi Nasional – Bandung, Indonesia 2023-2025
Internet of Things (IoT) Projects

- Designed and developed an automatic fan, flood detection device, and smart door lock.
- Developed a Mini RC Tank controlled through a mobile application and monitored through a web interface using MQTT.
- Developed an RC Tank arena with ArUco marker-based detection.

Organizational Experience

Itenas Club Reads – Bandung, Indonesia 2024-2025
Secretary

- Managed organizational administration, including meeting records, document archiving, and activity reports.
- Supported the planning and implementation of programs to promote reading activities within the campus community.

Unit Kegiatan Mahasiswa Itenas No Nippon Kurabu (INKU) – Bandung, Indonesia 2023-2025
Event Logistics Staff

- Managed the flow of equipment and materials from procurement and distribution to utilization by each division.
- Coordinated equipment and logistical requirements before, during, and after events.

Skills & Activities

- **Technical Skills**: Python, Java, PHP, JavaScript, Go, HTML & CSS, Laravel, Flask, Node.js, Spring Boot, Express.js, MySQL, AWS, Git/GitHub, Microsoft Office, Google Workspace.
- **System Design & Architecture:** Relational Database Design, System Workflow / Data Flow Diagram (DFD), Entity-Relationship Diagram (ERD), UML (Unified Modeling Language), RESTful API Design,
- **Machine Learning & Deep Learning**: ARIMA, K-Means, CNN, KNN, NLP, TF-IDF, Vector Space Model (VSM), YOLO.
- **Cloud & Infrastructure**: AWS EC2, S3, RDS, Lambda, API Gateway, SNS, DynamoDB, VPC, IAM, Cloud Architecture, Computer Networking, IP Addressing, Router, Switch, Access Point, MQTT.
- **Soft Skills**: Problem Solving, Teamwork, Mentoring, Communication, Leadership, Time Management, Adaptability, Organized.
- **Training** (2026): Huawei ICT Academy – HCIA-Security v4.0 Course.
- **Bootcamp** (2024): Codespace ITENAS – Backend Development.
- **Training** (2024): Codespace ITENAS – Backend Development Bootcamp.